Inscribed polytopes in curves and surfaces: a dimension count, a centring argument, and a computer-assisted certificate

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Abstract

We study four inscribed-shape problems: an equilateral triangle or a square in a Jordan curve, and a regular tetrahedron or a cube in a Jordan surface. We place them in families that continue to every dimension, and record seven results.

1. **No cubes in general.** A transversality argument shows that a generic smooth closed hypersurface in $\mathbb{R}^n$ contains no inscribed copy of any polytope with more vertices than the dimension $n(n+1)/2+1$ of the similarity group. In particular a generic smooth, strictly convex surface in $\mathbb{R}^3$ contains no inscribed cube (Theorem 1). A computer-assisted proof gives an explicit one: a cubic "egg" with rational coefficients and no inscribed cube (Theorem 1.3).
2. **Simplices in convex bodies.** An elementary "centring jig" argument shows that every convex body in $\mathbb{R}^n$ contains an inscribed regular simplex in every orientation that is *admissible* for it, and in every orientation when its boundary is C^1 (Theorems 2 and 3). For smooth bodies this is a special case of Gromov's theorem, but the proof uses only Brouwer's fixed-point theorem.
3. **Admissibility is needed.** Bodies with sharp corners can lose orientations: an isosceles triangle with apex below 60° has no inscribed equilateral triangle pointing at its apex (Proposition 5).
4. **A limit criterion for squares.** Non-shrinking squares on approximations give a square on the limit (Lemma 4).
5. **A computer-assisted proof.** Every Jordan curve whose polar radius stays within 0.15 of that of the ellipse $x^2+y^2/0.36=1$ has an inscribed square (Theorem 6), proved in interval arithmetic. This class contains nowhere-differentiable curves that are neither locally monotone nor symmetric.
6. **The same in 3D.** Every surface whose radius stays within 0.13 of that of a $1\times0.8\times0.6$ ellipsoid has an inscribed regular tetrahedron (Theorem 7), and within 0.06 in every orientation within 4.16° of a given one. Within 0.07 it has one in **every** orientation (Corollary 7.3), by a covering of all rotations with 12,784 certificates. This includes surfaces that are not C^1, outside Gromov's theorem.
7. **Numerical studies.** Four studies probe the open continuous cases:
   - on fractal curves and surfaces refined level by level, the largest inscribed squares and the tracked regular tetrahedra converge rather than shrink;
   - rectangles of 9 aspect ratios are found on all 9 test curves;

- exhaustive counts give an odd number of squares and an even number of 2:1 rectangles on every generic curve where the count settles.

Every numerical claim is reproducible from the repository, and every reported inscribed shape passes an independent check to a relative tolerance of 10^{-9}.

1. Introduction

The **inscribed square problem** (Toeplitz, 1911) asks whether every Jordan curve in the plane contains the four vertices of a square. It is open. Proved cases include:

- convex curves (Emch 1913);
- smooth curves (Schnirelmann 1929);
- locally monotone curves, which include all polygons (Stromquist 1989);
- curves made of two Lipschitz graphs with small constant (Tao 2017);
- curves close to a smooth curve relative to its curvature (Chambers 2022).

See Matschke's survey (2014) for the history. The **rectangular peg problem**, for rectangles of a prescribed aspect ratio, was solved for smooth curves by Greene and Lobb (2021). Vaughan showed that *some* rectangle always exists. The equilateral triangle, and indeed any triangle up to similarity, is inscribed in every Jordan curve (Meyerson 1980, Nielsen 1992). In $\mathbb{R}^3$, Gromov (1969) showed that every C^1 embedded sphere contains an inscribed regular tetrahedron, indeed any simplex, in every orientation. The continuous case is open. Regular octahedra are inscribed in smooth convex bodies (Makeev 2001), and cross-polytopes in smooth convex bodies in $\mathbb{R}^n$ for n an odd prime power (Karasev 2009). Centrally symmetric convex bodies in $\mathbb{R}^3$ contain inscribed cubes (Hausel, Makai and Szűcs 2000).

The problem map. A similar copy of a polytope in $\mathbb{R}^n$ has $D(n) = n(n+1)/2 + 1$ degrees of freedom, and each vertex on a hypersurface imposes one equation. So the *expected dimension* of the family of inscribed copies is $D(n) - m$ for a polytope with m vertices.

Polytope	n	m	$D(n) - m$	Status
Equilateral triangle	2	3	1	solved
Square, rectangle	2	4	0	open (continuous curves)
Regular k-gon, $k \geq 5$	2	k	< 0	fails generically (Thm 1)
Regular simplex	n	$n+1$	$n(n-1)/2$	open (continuous); Thms 2–3 for convex
Cross-polytope	n	$2n$	$(n-1)(n-2)/2$	open
Cube, hypercube	$n \geq 3$	2^n	< 0	**false** generically (Thm 1)

The square's "skeletal jig" (two perpendicular chords bisecting each other with equal lengths) generalises to the cross-polytope, not the hypercube. The tetrahedron problem generalises to the simplex. In every dimension

$n \geq 3$, the only regular polytopes with non-negative expected dimension are the simplex and the cross-polytope.

Methods. The companion pages find inscribed shapes with a *skeletal jig*: rays from a pivot, with the target shape inscribed exactly when suitable ray lengths agree. They combine a centring iteration (the "rattle"), Newton and Levenberg–Marquardt solves, and continuation across refinement levels. Every reported shape is checked independently: its pairwise vertex distances against the target's at the best-fitting scale, and the distance of each vertex to the curve or surface, both relative to the shape's size, must be below 10^{-9}.

Complete line-by-line proofs of Theorems 1–3, Lemma 4, Proposition 5 and Theorems 6–7 are in the appendix (the repository's `PROOFS.md`). Sections 2–5 state the results and give the key steps.

2. Generic non-existence

> **Theorem 1.** Let $P \subset \mathbb{R}^n$ be a polytope with $m > D(n)$ vertices. For a residual set of $f \in C^\infty(\mathbb{R}^n, \mathbb{R})$ (Whitney topology), the zero set $f^{-1}(0)$ contains no inscribed copy of P.

Key steps.

1. The similar copies of P form an embedded $D(n)$-dimensional submanifold C_P of the space of m-tuples of distinct points.
2. Put $W = C_P \times \{0\}$ in the multijet space $J^0_m(\mathbb{R}^n, \mathbb{R})$. It has codimension $nm + m - D(n)$, which is larger than nm, the dimension of the domain of the multijet $j^0_m f$.
3. By the multijet transversality theorem, for generic f the multijet is transversal to W, so it misses W entirely.

Openness of convexity and of the radial-graph condition near $f_0 = |x|^2 - 1$ then gives:

> **Corollary 1.1.** There are smooth, strictly convex surfaces in $\mathbb{R}^3$, arbitrarily C^∞-close to the round sphere, with no inscribed cube.

Symmetric surfaces have cubes, as Hausel–Makai–Szűcs predict, but they form a meagre set, so this does not conflict with Theorem 1. Theorem 1 does not name a surface. One can be named and checked:

> **Theorem 1.3.** The surface $x^2 + \frac{16}{25}y^2 + \frac{400}{289}z^2 + \frac{3}{20}(\frac{1}{2}x^3 + xyz + \frac{7}{10}y^2z + \frac{3}{10}z^3) = 1$ (inside $|p| \leq \frac{3}{2}$) is a smooth, strictly convex sphere with no inscribed cube.

The proof is computer-assisted, in interval arithmetic.

- First, the surface is checked to be a strictly convex sphere, with $|\nabla F| \geq 1.56$ and Hessian norm at most 3.40.

- Taylor's theorem then shows that a cube with all its vertices on it would be nearly flat unless its edge is at least 0.153.
- A branch-and-bound search over centre, size and orientation, with the orientations reduced by the cube's 24 rotations, excludes every other cube: 63 million boxes, under a minute.
- An independent re-implementation with Arb's ball arithmetic agrees. In a control run on the ellipsoid (asymmetry 0), the search correctly finds that surface's cube.

The same programs show that the egg has no regular icosahedron. It has no regular dodecahedron either, since a dodecahedron's vertices include a cube's (Corollary 1.4). The proof also survives making the asymmetry as small as 10^{-8}, at almost no extra cost. So a surface within about 10^{-8} of an ellipsoid with an inscribed cube can have none.

3. Simplices in convex bodies: the centring jig

Let Δ be a simplex whose circumcentre is interior, $w_0, \dots, w_n$ the unit vectors from the circumcentre to its vertices, and $\lambda_k > 0$ with $\sum \lambda_k w_k = 0$. The vector λ spans the space of linear relations among the w_k. For a convex body K, let $r_k(x)$ be the length of the k-th arm of the jig at x, i.e. how far $x + t w_k$ stays in K. The **rattle field** is

$$V(x) = \sum_k \lambda_k \, r_k(x) \, w_k .$$

A zero of V forces $\lambda_k r_k = \rho \lambda_k$, i.e. equal arms, which is an inscribed copy of Δ.

> **Theorem 2.** If K is strictly convex with C^1 boundary, V has a zero in the interior. Hence K contains an inscribed copy of Δ, in particular a regular simplex, in every orientation.

Key steps.

1. **Continuous arms.** Each r_k is continuous; strict convexity is used here.
2. **Inward at the boundary.** $\langle V(p), \nu \rangle < 0$ at every boundary point; the C^1 boundary is used here.
3. **Zero.** An inward-pointing continuous field on a convex body has a zero, by Brouwer applied to $x \mapsto \pi_K(x + V(x))$.

> **Theorem 3.** Let K be any convex body. Call an orientation *admissible* if from every boundary point some arm of the jig points into the interior of the tangent cone. Then K contains an inscribed copy of Δ in every admissible orientation. If ∂K is C^1, every orientation is admissible.

Key steps. Apply Theorem 2 to smooth, strictly convex outer approximations $K \subset K_m \subset K + \frac{3}{m} B$ (Schneider). Admissibility prevents the inscribed simplices from shrinking to a boundary point. A limit of the non-shrinking simplices is inscribed in K.

> **Proposition 5.** An isosceles triangle with apex angle below 60° has no inscribed equilateral triangle with a vertex pointing at the apex along the axis. The bound 60° is sharp.

So "every orientation" genuinely fails for non-smooth bodies, which is the phenomenon Gromov's C^1 hypothesis rules out. In the plane some orientation always works (Meyerson).

Why the square needs more. A square's jig has a two-dimensional relation space ($w_0 + w_2 = 0$, $w_1 + w_3 = 0$). So the centring step yields only a rhombus, and the equal-diagonal condition requires varying the rotation. That intermediate-value step is where non-convex curves break the argument. The same happens for the cross-polytope in every dimension.

4. A limit criterion and what counting cannot do

> **Lemma 4.** Let $\gamma_L \to \gamma$ uniformly, with γ a Jordan curve. Suppose infinitely many γ_L have inscribed squares whose opposite vertices lie at parameter distance at least $\delta > 0$. Then γ has an inscribed square.

For generic smooth curves the number of inscribed squares is odd, and the number of rectangles of any other fixed aspect ratio is even. Section 6.3 confirms both numerically. On rough curves small squares multiply at every scale, and in the limit their number is infinite. So a proof for continuous curves has to control the *sizes* of squares, as in Lemma 4, not their number.

5. Computer-assisted certificates

Let E be the ellipse with semi-axes 1 and $b = 0.6$, with polar radius $r_E(\theta) = (\cos^2\theta + \sin^2\theta / b^2)^{-1/2}$. For continuous $\rho > 0$ let $\gamma_\rho(\theta) = \rho(\theta)(\cos\theta, \sin\theta)$.

> **Theorem 6.** If $|\rho(\theta) - r_E(\theta)| \le 0.15$ for all θ, then γ_ρ has an inscribed square, with diagonal at least 0.754.

Proof outline.

1. **Square equations.** With A and C as opposite corners, the other corners are
$$B, D = \tfrac{1}{2}\big((I \mp R)A + (I \pm R)C\big),$$
where R is the rotation by 90°. Parametrise A and C by the rational parametrisation $\gamma_E(t) = \big(\frac{1-t^2}{1+t^2}, \frac{2bt}{1+t^2}\big)$ and its antipode, carried along rays to the perturbed curve. Let $G_\rho(t, t') = (\mathrm{sd}_\rho(B), \mathrm{sd}_\rho(D))$, where sd_ρ is the signed distance to γ_ρ.

2. **Perturbation bound.** For radial curves, $|\rho - \sigma| \le T$ implies $|\mathrm{sd}_\rho - \mathrm{sd}_\sigma| \le T$ everywhere: compare along rays when the two signs differ. The corners B, D move by at most $\sqrt{2}\,T$. So $\|G_\rho - G_E\|_\infty \le (1+\sqrt{2})T$.

3. **Degree lemma.** Let $Q = z + K([-c,c]^n)$ for any point z and matrices J, K with $\det J \neq 0$, and let $L(p) = J(p - z)$, so $\|L\|_\infty \ge c_{\text{eff}} = c(1 - \|JK - I\|_\infty) > 0$ on ∂Q. Suppose that on ∂Q:

   - $\|G_E - L\|_\infty < \|G_E\|_\infty$, which makes the homotopy from L to G_E zero-free, so $\deg(G_E, Q, 0) = \operatorname{sign}\det J$;
   - and $\tau < \|G_E\|_\infty$.

   Then every G within τ of G_E has a zero in Q. This test needs no winding number and works in any dimension.

4. **The computation**, all in interval arithmetic with outward rounding:

   - on ∂Q, from 80,000 samples with $c = 0.5$ and explicit Lipschitz bounds between them, $\|G_E\|_\infty \ge 0.3660 > (1+\sqrt{2}) \cdot 0.15 = 0.3621$;
   - and $\|G_E\|_\infty - \|G_E - L\|_\infty \ge 0.23 > 0$;
   - and $|A - C| \ge 1.054$ on Q, so the square is non-degenerate.

   The signed distance to the ellipse is enclosed exactly: the closest-point equation's root is bracketed and the bracket verified in intervals. No trigonometric functions are used.

The computation takes under a second (`node tools/certify-square.js`), and runs live on the companion page. An independent re-implementation in Python, using the interval arithmetic of the mpmath library and sharing only the box definition, confirms it (`tools/verify_certificates.py`), as does a third in C with the Arb library (`tools/verify-certificates.c`). Only IEEE 754 arithmetic ($+, -, \times, \div, \sqrt{}$, each correctly rounded) and about 300 lines of code are trusted.

> **Corollary 6.1.** Let $W(\theta) = (1 - 2^{-1/2}) \sum_{k \ge 0} 2^{-k/2} \cos(2^k(\theta - 0.3))$. The curve $\rho = r_E + 0.15\,W$ has an inscribed square, and so does $r_E + 0.15 f$ for every continuous $|f| \le 1$.

The curve $r_E + 0.15\,W$ lies outside the classical cases:

- it is nowhere differentiable (Hardy);
- it is not locally a graph in any direction, since a monotone coordinate would make W differentiable almost everywhere, so it is not locally monotone;
- it has no reflection or point symmetry.

Theorem 6's neighbourhood is 4.2 times wider than $1/(10\kappa) = 0.036$ for this ellipse. We do not claim that no other smooth reference curve brings Corollary 6.1 within Chambers' theorem. The certificate's value is that it is explicit, checkable, and reaches a rough class directly.

Tetrahedra on rough surfaces. In 3D, fix the tetrahedron's orientation, with unit vectors $w_0, \dots, w_3$ from its centre to its vertices; an exactly orthogonal rational rotation (a Cayley transform) keeps them exact. The unknowns are the centre x and circumradius s, and the equations are $G(x, s) = (\mathrm{sd}(x + s w_k))_k \in \mathbb{R}^4$. The vertices do not depend on the surface, so a radial perturbation of size T moves G by at most T, and the degree lemma applies with $n = 4$.

> **Theorem 7.** Let E be the ellipsoid with semi-axes $1, 0.8, 0.6$. Every surface $\rho(\omega)\omega$ with $|\rho - r_E| \le 0.13$ has an inscribed regular tetrahedron (in a fixed orientation) with circumradius at least 0.4089. If $|\rho - r_E| \le 0.06$, it has one in every orientation within 4.16° of that one.

The computation (`node tools/certify-tetra.js` , seconds on 4 cores) checks the 8 faces of the 4D box ($c = 0.26$) in adaptive cells. Each cell is checked at its centre, with an explicit bound for the change across it, and split only if the margin is too small for its size. All 997,168 cells pass both conditions of the degree lemma with $\tau = T = 0.13$. A uniform grid would need 64 million samples for the weaker $T = 0.12$. An independent Python/mpmath implementation reaches the same 997,168 cell decisions.

Turning the tetrahedron by an angle θ moves G by at most $s_{\max}\theta$, which acts as one more perturbation. That gives the open set of orientations.

> **Corollary 7.3.** If $|\rho - r_E| \le 0.07$, the surface has an inscribed regular tetrahedron in every orientation.

Write rotations as unit quaternions, with Cayley vector $q = v/w$. The tetrahedron's symmetries are the 24 Hurwitz units, and every rotation is equivalent under them to one whose Cayley vector lies in the octahedron $|a| + |b| + |c| \le 1$. That octahedron is covered by 10,544 cubes of half-width $1/36$, and the 320 where the room is tight are split into 8. Within a cube of half-width h, the vertices turn by at most $2\sqrt{3}\, h\, s/d$, where d is the smallest value of $|(1, q)|$ on the cube. So the certificate at each cube's centre, with that turning counted as an extra perturbation, covers the whole cube. All 12,784 certificates pass (16.3 billion cells, about 4 hours on 4 cores). The covering is re-checked in exact arithmetic, and a random sample of 100 cubes in C with Arb.

With $F(\omega) = (1 - 2^{-1/2}) \sum_k 2^{-k/2} \cos(2^k \langle \omega, e \rangle)$, the surface $r_E + 0.13\, F$ is not C^1, so Gromov's theorem does not cover it, but Theorem 7 does.

A negative result. The same method fails on a self-similar Weierstrass blob $\rho = 1 + \sum_k 0.085 \cdot 2^{-k/2} \cos(m 2^k \theta + \phi_k)$. Taking the K-octave truncation as the base curve, the certifiable perturbation is about $1/5$ of the remaining tail at every K tested:

K	tail	certifiable
2	0.145	0.026
3	0.103	0.023
4	0.073	0.014

Each octave shrinks both quantities by the same factor. A certificate of this kind needs a base curve whose square is robust at a scale larger than the remaining roughness. A curve with self-similar roughness at a fixed relative size never provides one. This is a quantitative form of the difficulty of the open case.

6. Numerical studies

All studies run in the browser (`pages/inscribed-study.html`) and from the command line (`node tests/run-study.js all`). Raw data are in `data/` .

6.1 Square shrinkage on fractal curves

Four fractal families are refined level by level with a fixed seed, so the levels converge uniformly to a limit curve. At each level, squares are continued from the previous level and searched for afresh.

Family, roughest level	Edges	Largest square / width
Koch snowflake, level 7	49,152	0.910 (unchanged since level 4)
Minkowski island, level 5	131,072	1.017
Weierstrass blob, level 14	131,072	0.829 (steady from level 11)
Random coastline, level 16	65,536	0.832 (0.83–0.85 since level 10)

The tracked squares change size by less than 1% per level at the highest levels. This is the hypothesis of Lemma 4, observed for finitely many levels.

6.2 Tetrahedra on fractal surfaces and rectangles of every aspect ratio

Sixteen regular tetrahedra, started at fixed orientations, are followed on three fractal surface families up to 16 octaves:

- Weierstrass-type, Hölder exponent 0.5;
- ridged;
- Hölder exponent 0.3.

The smallest tracked tetrahedron changes from 0.895 to 0.894, 0.899 to 0.948, and 0.939 to 0.927 (circumradius divided by mean radius) between levels 8 and 16. Some tetrahedra are lost along the way, through pair annihilation or tracking failure.

Rectangles with aspect ratios from $1{:}1$ to $16{:}1$ were found on all 9 test curves, in all 81 cells. On the ellipse, the search finds exactly one square and two rectangles of every other ratio.

6.3 Parity

A (u, v) grid search with Newton refinement, repeated on finer grids until the count stabilises, enumerates every inscribed square and 2:1 rectangle. Over 44 curves and levels:

- the square count settled in 38 cases and was odd in all of them, from 1 up to 171;
- the rectangle count settled in 34 cases and was even in all of them.

Signed counts vanish, because a square's two diagonals carry opposite signs. Counts fail to settle on non-generic curves (circles, long parallel edges) and on rough curves, where small squares appear at every scale.

7. Open problems

Problem	Status
Square in every Jordan curve	open
Rectangle of every aspect ratio in every Jordan curve	open; smooth case solved
Regular simplex in every Jordan sphere	open; C^1 case solved (Gromov); convex case in admissible orientations, and a neighbourhood of one ellipsoid, in every orientation within 0.07 (Thm 7, Cor 7.3), here
Cross-polytope in every (smooth convex) sphere	open in general
Cube in every Jordan surface	false (Theorem 1), with an explicit egg (Thm 1.3); true for centrally symmetric convex bodies

Two directions follow from this work:

- **Larger certified neighbourhoods.** Theorems 6–7 could start from other smooth base curves and surfaces, use several boxes, or replace the linearisation by a higher-order model.
- **Wider 3D certificates.** Second-derivative bounds for the signed distance would enlarge Theorem 7's neighbourhood, and a finer covering of the rotations would give every orientation for a wider neighbourhood than 0.07.

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Appendix: complete proofs

This appendix is the repository file `PROOFS.md`, reproduced verbatim from its notation section on.

Notation

- A **similarity** of $\mathbb{R}^n$ is a map $g(x) = sQx + c$ with $s > 0$, $Q \in O(n)$, $c \in \mathbb{R}^n$. The similarities form a Lie group $\mathrm{Sim}(n)$ of dimension

$$D(n) = \underbrace{n}_{\text{translation}} + \underbrace{\tfrac{n(n-1)}{2}}_{\text{rotation}} + \underbrace{1}_{\text{scale}} = \tfrac{n(n+1)}{2} + 1.$$

  So $D(2) = 4$, $D(3) = 7$, $D(4) = 11$, $D(5) = 16$.
- Let P be a polytope in $\mathbb{R}^n$ with vertices $v_1, \dots, v_m$ that affinely span $\mathbb{R}^n$. A copy of P is **inscribed** in a set Σ if $g(v_i) \in \Sigma$ for all i for some $g \in \mathrm{Sim}(n)$. Since $s > 0$, the copy is non-degenerate.
- The **expected dimension** of the family of inscribed copies of P in a hypersurface is $D(n) - m$: each vertex lying on Σ is one equation.

Polytope	n	Vertices m	$D(n) - m$
Equilateral triangle	2	3	1
Square	2	4	0
Regular pentagon, hexagon, …	2	5, 6, …	−1, −2, …
Regular tetrahedron	3	4	3
Regular octahedron	3	6	1
Cube	3	8	**−1**
Icosahedron, dodecahedron	3	12, 20	−5, −13
Regular simplex	n	$n+1$	$n(n-1)/2 \geq 0$
Cross-polytope (orthoplex)	n	$2n$	$(n-1)(n-2)/2 \geq 0$
Hypercube	$n \geq 3$	2^n	**negative**

In every dimension $n \geq 3$, the only regular polytopes with non-negative expected dimension are the simplex and the cross-polytope. (The 24-, 120- and 600-cells in $\mathbb{R}^4$ have 24, 600 and 120 vertices, all more than $D(4) = 11$.)

Theorem 1 — generic non-existence

> **Theorem 1.** Let $P \subset \mathbb{R}^n$ be a polytope with $m > D(n)$ vertices. There is a residual set $\mathcal{R} \subset C^\infty(\mathbb{R}^n, \mathbb{R})$ (Whitney C^∞ topology) such that for every $f \in \mathcal{R}$, the zero set $f^{-1}(0)$ contains no inscribed copy of P.

Tool. *Multijet Transversality Theorem* (Golubitsky–Guillemin, *Stable Mappings and Their Singularities*, Ch. II, Thm. 4.13). Let X, Y be smooth manifolds and W a submanifold of the s-fold k-jet bundle $J^k_s(X,Y)$. Then $\{f \in C^\infty(X,Y) : j^k_s f \pitchfork W\}$ is residual in the Whitney C^∞ topology.

We use $k = 0$. Let $X^{(m)} \subset (\mathbb{R}^n)^m$ be the open set of m-tuples of **distinct** points. Then $J^0_m(\mathbb{R}^n, \mathbb{R}) = X^{(m)} \times \mathbb{R}^m$, and

$$j^0_m f(x_1, \dots, x_m) = \big(x_1, \dots, x_m,\ f(x_1), \dots, f(x_m)\big).$$

Step 1: copies of P form a submanifold of dimension $D(n)$. Define $\Phi : \mathrm{Sim}(n) \to X^{(m)}$, $\Phi(g) = (gv_1, \dots, gv_m)$. The points are distinct because g is injective.

- *Injective.* If $\Phi(g) = \Phi(h)$, the affine map $h^{-1}g$ fixes every v_i. The v_i affinely span $\mathbb{R}^n$, so $h^{-1}g = \mathrm{id}$.
- *Immersion.* Φ is the orbit map $g \mapsto g \cdot \Phi(\mathrm{id})$ of the smooth diagonal action of $\mathrm{Sim}(n)$ on $X^{(m)}$. Orbit maps have constant rank, and an injective constant-rank map is an immersion (constant rank theorem).
- *Homeomorphism onto its image.* Choose $n+1$ affinely independent vertices $v_{i_0}, \dots, v_{i_n}$. Given $x = \Phi(g)$, g is the unique affine map with $g(v_{i_j}) = x_{i_j}$. Solving for it is linear algebra, so g depends

continuously on x.

So $C_P := \Phi(\mathrm{Sim}(n))$ is an embedded submanifold of $X^{(m)}$ of dimension $D(n)$.

Step 2: the bad set. Let $W = C_P \times \{0\} \subset X^{(m)} \times \mathbb{R}^m$. It is a submanifold of dimension $D(n)$ in a manifold of dimension $nm + m$, so

$$\operatorname{codim} W = nm + m - D(n).$$

Step 3: transversality forces emptiness. For f in the residual set given by the Multijet Transversality Theorem, $j^0_m f \pitchfork W$. The domain of $j^0_m f$ has dimension nm. If $nm < \operatorname{codim} W$, which is exactly $m > D(n)$, a transversal map cannot meet W at all: at an intersection point the image of the differential (dimension $\le nm$) plus TW would have to span a space of dimension $nm + m$, which is impossible.

Step 4: conclusion. Suppose $f^{-1}(0)$ contained an inscribed copy $g(P)$. Then $x = \Phi(g) \in C_P$ and $f(x_i) = 0$ for all i, so $j^0_m f(x) \in W$, contradicting Step 3. ■

> **Corollary 1.1 (no inscribed cube in general).** There exist smooth, strictly convex surfaces in $\mathbb{R}^3$, arbitrarily close in C^∞ to the round sphere, that contain no inscribed cube. Hence the conjecture "every closed Jordan surface contains an inscribed cube" is false.

Proof. Let $f_0(x) = |x|^2 - 1$, whose zero set is the unit sphere. The set $\mathcal{U}$ of f satisfying all of the following is open in the Whitney C^2 topology, hence in C^∞, and contains f_0:

1. $f < 0$ on a small ball around the origin, and $f > 0$ outside a large ball;
2. $x \cdot \nabla f(x) > 0$ on the closed annulus in between;
3. wherever $f = 0$, the Hessian of f is positive definite on $\ker \nabla f$.

(For f_0, take the balls of radius $1/2$ and 2: then $x \cdot \nabla f_0 = 2|x|^2 > 0$ on the annulus, and the Hessian is $2I$.) For $f \in \mathcal{U}$, conditions 1 and 2 make each ray from the origin cross $f^{-1}(0)$ exactly once and transversally, so $f^{-1}(0)$ is a smooth radial graph, a smooth sphere. Condition 3 gives it a positive definite second fundamental form, so it bounds a strictly convex body (Hadamard). $C^\infty(\mathbb{R}^3)$ with the Whitney topology is a Baire space, so the residual set $\mathcal{R}$ of Theorem 1 (with P the cube, $m = 8 > 7 = D(3)$) is dense, and $\mathcal{R} \cap \mathcal{U}$ contains points arbitrarily close to f_0. ■

Corollary 1.2. By the same argument, generic smooth Jordan curves contain no inscribed regular k-gon for $k \ge 5$ ($k > D(2) = 4$). Generic smooth closed hypersurfaces in $\mathbb{R}^n$, $n \ge 3$, contain no inscribed hypercube ($2^n > D(n)$). And generic smooth closed surfaces in $\mathbb{R}^3$ contain no inscribed icosahedron or dodecahedron.

Theorem 1.3 (an explicit egg with no inscribed cube). Let

$$F(x,y,z) = x^2 + \tfrac{16}{25}y^2 + \tfrac{400}{289}z^2 + \tfrac{3}{20}\left(\tfrac{1}{2}x^3 + xyz + \tfrac{7}{10}y^2 z + \tfrac{3}{10}z^3\right) - 1$$

and $S = \{p : |p| \le \tfrac{3}{2},\ F(p) = 0\}$, the repository's *lopsided egg* (semi-axes about 1, $\tfrac{5}{4}$, $\tfrac{17}{20}$, with no symmetry). Then S is a smooth sphere bounding a strictly convex body, and no cube has all eight vertices on S.

So the cube analogue of the square peg problem fails for an explicit, smooth, strictly convex surface, and not only for unnamed generic ones.

Proof. Computer-assisted, in three parts (`tools/egg-cube.c` ; parts 1–2 take about a second, part 3 about 40 seconds on 4 cores). All bounds use interval arithmetic with outward rounding.

- **Part 1: the surface.** $[-\frac{3}{2}, \frac{3}{2}]^3$ is split into boxes. On every box that meets the sphere $|p| = \frac{3}{2}$, $F > 0$. On every box inside the ball where F may vanish, $p \cdot \nabla F > 0$.
  - Along each ray from the origin, F starts at $F(0) = -1$, ends positive, and crosses zero only upwards, so it vanishes exactly once. Hence S is a star-shaped sphere, smooth because $\nabla F \neq 0$ on it.
  - The same boxes give $0.8329 \leq |p| \leq 1.2630$ and $|\nabla F| \geq g = 1.5612$ on S, and a bounding box for S.
  - The Hessian of F is positive definite on the ball $|p| \leq 1.2630$ (all leading minors positive on boxes covering it). So F is strictly convex there, the body $\{F \leq 0\}$ inside that ball is strictly convex, and S, its boundary, has positive curvature. The Hessian's norm there is at most $M = 3.3939$.
- **Part 2: no small cubes.** Let a cube of edge a have its vertices $v_0, \ldots, v_7$ on S, and let $n = \nabla F(v_0)/|\nabla F(v_0)|$.
  - Each segment $v_0 v_i$ lies in the ball, so by Taylor's theorem $0 = F(v_i) - F(v_0) = \nabla F(v_0) \cdot d + \frac{1}{2} d^{\mathsf{T}} H d$ with $d = v_i - v_0$. Hence $|n \cdot d| \leq M|d|^2/(2g) \leq 3Ma^2/(2g)$, using $|d| \leq \sqrt{3}\, a$.
  - So all eight vertices lie in a slab of thickness $3Ma^2/g$. A cube is at least a wide in every direction, so $a \leq 3Ma^2/g$, and $a \geq g/(3M) = 0.1533$.
  - At the other end, the cube's diameter $2\sqrt{3}\, s$ (with $s = a/2$) is at most S's diameter, and its centre, the average of its vertices, lies in S's bounding box.
- **Part 3: no other cubes.** Write a cube as centre c, half-edge $s \in [0.0767, 0.7292]$ and rotation $R(q)$.
  - **Orientations.** The cube's 24 rotations are the 48 units of the binary octahedral group: $\pm 1, \pm i, \pm j, \pm k$, $\frac{1}{2}(\pm 1 \pm i \pm j \pm k)$, and $(\pm a \pm b)/\sqrt{2}$ for distinct $a, b \in \{1, i, j, k\}$. As in Corollary 7.3, every rotation is equivalent to one closest to 1 among these units. Its Cayley vector satisfies $|q_i| \leq \sqrt{2} - 1$ (from $(1 \pm i)/\sqrt{2}$ and the like) and $|q_1| + |q_2| + |q_3| \leq 1$ (from the Hurwitz units).
  - **The search.** The 7-dimensional region of (c, s, q) is searched by branch and bound. A box is excluded if, for some vertex $c + sR(q)e$, $|F| > G\rho$ at the box's centre. Here $\rho = |\Delta c| + \sqrt{3}\,|\Delta s| + 2\sqrt{3}\, s_{\max} |\Delta q|$ bounds how far that vertex moves across the box (as in Corollary 7.3), and G bounds $|\nabla F|$ within ρ of it. Otherwise the box is split.
  - **Result.** All 63,130,228 boxes are excluded, and none is split more than 37 times. ■

Checks.

- **Control.** With the asymmetry set to 0 (`-DE_NUM=0`), the surface is an ellipsoid. That is centrally symmetric, so it has inscribed cubes (Hausel–Makai–Szűcs). The search then fails exactly where it

should: around the cube with centre 0 and half-edge $1/\sqrt{1+16/25+400/289} = 0.57505$. So the program finds cubes when they exist.

- **Independent check** (`tools/egg-cube-arb.c` , about 10 minutes). A second program, in C with Arb's ball arithmetic, shares no code with the first. It bounds boxes differently: it evaluates the vertices and F directly on balls of parameters, instead of using a Lipschitz bound. It redoes all three parts: 367 million boxes, all excluded. Its own control (`./egg-cube-arb control`) finds the ellipsoid's cube.
- **Margin.** The method has room to spare. Replace the asymmetry $\frac{3}{20}$ by smaller ε, which moves the surface by at most about ε towards the ellipsoid, a surface that *does* have a cube. The proof still goes through, and its cost grows only like $\log(1/\varepsilon)$ (`-DE_NUM=1 -DE_DEN=...`):

ε	$\frac{3}{20}$	10^{-2}	10^{-3}	10^{-4}	10^{-6}	10^{-8}
boxes (millions)	63.1	51.7	54.8	58.7	66.6	74.5
deepest split	37	66	87	112	157	205
seconds (4 cores)	45	32	34	37	42	46

So a surface within about 10^{-8} of an ellipsoid with an inscribed cube has none. That is the instability Theorem 1 predicts: the ellipsoid's cube exists only because of its symmetry. Much below 10^{-8}, the boxes would have to be smaller than the program's floor of 10^{-9}, which is set by double precision.

Corollary 1.4 (more shapes on the egg). The egg S of Theorem 1.3 also contains no inscribed regular icosahedron or dodecahedron, and no inscribed copy of any polytope whose vertices include those of a cube (for example the rhombic dodecahedron).

Proof.

- **Shapes containing a cube.** Any copy of such a polytope would contain an inscribed cube. The regular dodecahedron is one: its 20 vertices include the 8 of a cube, $(\pm 1, \pm 1, \pm 1)$ in the usual coordinates.
- **The icosahedron.** Its vertices $(0, \pm 1, \pm\varphi)$ and their cyclic permutations include no cube, so it needs its own search (`./egg-cube icosahedron` , about 2 minutes).
  - The icosahedron's rotations include the tetrahedral ones (the Hurwitz units of Corollary 7.3), so orientations with $|q_1| + |q_2| + |q_3| \le 1$ suffice.
  - The small-shape lemma holds for any polytope. Let W be its minimal width and D its diameter. All vertices lie in a slab of thickness MD^2/g, so $W \le MD^2/g$. For the icosahedron at scale s, $W = 2\varphi^2 s/\sqrt{3}$ and $D = 2\sqrt{\varphi+2}\,s$, which gives $s \ge 0.0961$.
  - The branch and bound excludes all 129 million boxes.
- **Checks.** The Arb program (`./egg-cube-arb icosahedron`) re-checks the icosahedron independently: 254 million boxes, about 9 minutes. As a control, both programs, run on the unit sphere (`-DE_NUM=0 -DSPHERE` , and `./egg-cube-arb icosahedron-control`), find its icosahedra at scale $1/\sqrt{\varphi+2} = 0.5257$. The ellipsoid cannot serve as the control here: the only quadric through the 12 vertices of a regular icosahedron is its circumsphere, so an ellipsoid that is not a sphere has none. ■

In 4D (`tools/egg-tesseract.c`), the analogous hypersurface $F + \frac{100}{81}w^2 + \frac{3}{20}(xzw + \frac{2}{5}w^3) = 0$ is proved to be a smooth, strictly convex 3-sphere. Its slice $w = 0$ is the egg S. The same lemma confines any inscribed tesseract to half-edge $s \in [0.0525, 0.6425]$. Whether it has an inscribed tesseract is **not settled**: the branch and bound now runs over 11 parameters, and a random-probe estimate puts it at about 10^{14} boxes. The closest tesseract a numerical search finds is off by 3.3% of its edge. The Dimension Explorer shows both eggs with the closest cube and tesseract.

In the plane the analogous question is classical. A regular pentagon's five vertices lie on its circumcircle, and five points determine a conic. So no ellipse other than a circle contains a regular pentagon, and no computation is needed.

Remarks.

- Theorem 1 is non-constructive: it shows that counterexamples are everywhere, but not where. Theorem 1.3 gives one explicitly. Before it was proved, two independent solvers found no cube on the egg closer than 1.5–2.8% of its size.
- **Symmetry is not generic.** A centrally symmetric surface has inscribed cubes (Hausel–Makai–Szűcs 2000), and so, numerically, do mirror-symmetric ones. Symmetric surfaces form a small (meagre) set, so this does not contradict Theorem 1.
- Theorem 1 says nothing about shapes with $m \le D(n)$. For those, the count only says solutions are *expected*; existence needs a separate argument, such as Theorem 2.

Theorem 2 — the centring jig

> **Theorem 2.** Let $K \subset \mathbb{R}^n$ be a compact convex body (non-empty interior) that is **strictly convex** and has a C^1 **boundary**. Let Δ be a simplex whose circumcentre lies in its interior, and let $w_0, \dots, w_n$ be the unit vectors from that circumcentre to the vertices. Then for every $Q \in O(n)$ there are $x^* \in \operatorname{int} K$ and $\rho > 0$ such that the points $x^* + \rho\, Q w_k$ ($k = 0, \dots, n$) all lie on ∂K. These are the vertices of a copy of Δ inscribed in ∂K, with the prescribed orientation Q.
>
> In particular, K contains an inscribed **regular** simplex (an equilateral triangle in 2D, a regular tetrahedron in 3D) in every orientation.

Replacing w_k by Qw_k, we may assume $Q = I$. Because the circumcentre is interior, there are weights $\lambda_k > 0$ with

$$\textstyle\sum_k \lambda_k w_k = 0. \qquad (*)$$

The w_k span $\mathbb{R}^n$ (they point to the vertices of a non-degenerate simplex from an interior point), so the linear relations among $w_0, \dots, w_n$ form a one-dimensional space, spanned by $\lambda = (\lambda_0, \dots, \lambda_n)$. For a regular simplex, $\lambda_k = 1$.

The jig. For $x \in K$ let $r_k(x) = \max\{t \ge 0 : x + t w_k \in K\}$, the length of the jig's k-th arm. It is finite because K is compact. The **rattle field** is

$$V(x) = \sum_{k} \lambda_k \, r_k(x) \, w_k .$$

For the regular simplex, $V(x)/(n+1)$ is the vector from x to the centroid of the $n+1$ hit points. The finders' "rattle" $x \leftarrow x + \alpha V(x)$ is the iteration that looks for a zero of V.

Lemma A (the arms vary continuously). Each r_k is continuous on K.

Proof. Write $w = w_k$, $r = r_k$.

- *Upper semicontinuity:* if $x_j \to x$ and $r(x_j) \to t$, then $x_j + r(x_j) w \in K$, and K is closed, so $x + t w \in K$ and $r(x) \ge t$.
- *Lower semicontinuity:* let $0 < t < r(x)$. The points x and $x + r(x) w$ lie in K. If $x \in \operatorname{int} K$, the half-open segment from an interior point to a point of a convex set lies in the interior, so $x + t w \in \operatorname{int} K$. If $x \in \partial K$, the two points are distinct points of K, and strict convexity puts the open segment between them in $\operatorname{int} K$; again $x + t w \in \operatorname{int} K$.

  So for $x_j \to x$, eventually $x_j + t w \in K$, i.e. $r(x_j) \ge t$. Hence $\liminf r(x_j) \ge r(x)$. (If $r(x) = 0$ there is nothing to prove.) $\blacksquare$

Lemma B (the rattle points inward at the boundary). For $p \in \partial K$ with outward unit normal ν (unique, since ∂K is C^1), $\langle V(p), \nu \rangle < 0$.

Proof. $K \subset \{y : \langle y - p, \nu \rangle \le 0\}$.

- If $\langle w_k, \nu \rangle > 0$, then $p + t w_k \notin K$ for every $t > 0$, so $r_k(p) = 0$.
- If $\langle w_k, \nu \rangle < 0$, then $r_k(p) > 0$. Since the supporting hyperplane at p is unique, the tangent cone $T = \overline{\bigcup_{t>0} t(K - p)}$ is the half-space $\{d : \langle d, \nu \rangle \le 0\}$. The convex cone $C = \bigcup_{t>0} t(K-p)$ has closure T, and a convex set has the same interior as its closure. So $w_k \in \operatorname{int} T = \operatorname{int} C \subset C$, meaning $p + s w_k \in K$ for some $s > 0$.

By $(*)$, $\sum_k \lambda_k \langle w_k, \nu \rangle = 0$. Not all $\langle w_k, \nu \rangle$ are zero, because the w_k span $\mathbb{R}^n$. So some $\langle w_k, \nu \rangle < 0$. Therefore

$$\langle V(p), \nu \rangle = \sum_{\langle w_k, \nu \rangle < 0} \lambda_k \, r_k(p) \, \langle w_k, \nu \rangle < 0,$$

since the terms with $\langle w_k, \nu \rangle > 0$ vanish ($r_k(p) = 0$) and at least one remaining term is strictly negative. $\blacksquare$

Lemma C (an inward field has a zero). Let K be a compact convex body and $V : K \to \mathbb{R}^n$ continuous with $\langle V(p), \nu \rangle < 0$ for every $p \in \partial K$ and every outward unit normal ν at p. Then $V(x^*) = 0$ for some $x^* \in \operatorname{int} K$.

Proof. Let π_K be the nearest-point projection onto K (continuous). The map $\varphi(x) = \pi_K(x + V(x))$ sends K continuously into K, which is homeomorphic to a closed ball. By Brouwer's fixed-point theorem, $\varphi(x^*) = x^*$ for some x^*. Let $y = x^* + V(x^*)$.

- If $y \in K$, then $\pi_K(y) = y$, so $V(x^*) = 0$.
- Otherwise $x^* = \pi_K(y) \in \partial K$, and $y - x^* = V(x^*)$ is a non-zero outward normal vector at x^*: $V(x^*) = \mu\nu$ for an outward unit normal ν and $\mu > 0$. Then $\langle V(x^*), \nu \rangle = \mu > 0$, a contradiction.

So $V(x^*) = 0$, and $x^* \notin \partial K$ because $V \neq 0$ there. ■

Proof of Theorem 2. By Lemmas A and B, V is continuous and points inward on ∂K. By Lemma C it has a zero $x^* \in \operatorname{int} K$: $\sum_k \lambda_k r_k(x^*) w_k = 0$. So $(\lambda_k r_k(x^*))_k$ is a linear relation among the w_k, hence a multiple of λ: $\lambda_k r_k(x^*) = \rho\lambda_k$ for all k, i.e. **all arms have the same length** ρ.

Since x^* is interior, $\rho > 0$. Each point $x^* + \rho w_k$ is in K (closedness), while $x^* + t w_k \notin K$ for $t > \rho$, so it lies on ∂K. These $n + 1$ points lie on the sphere of radius ρ about x^* in the directions w_k, so they are the vertices of $\rho\Delta$ translated to x^*. ■

Remarks.

- *Every orientation.* The family of solutions is parametrised by $Q \in O(n)$, of dimension $n(n-1)/2$. This matches the expected dimension $D(n) - (n+1)$ in the table.
- *Pivot only.* The finders' **Pivot only** mode is exactly Theorem 2: it fixes Q and searches only over x.
- *Where it stops.* Strict convexity is used in Lemma A, and C^1 in Lemma B. For convex bodies with corners or flat pieces, and for non-convex or merely continuous surfaces, Lemmas A and B can fail: the arm lengths can jump, which the 2D finder counts as "discontinuities". Gromov's theorem removes convexity in the C^1 case; the merely continuous case is open.
- *Why the square needs more.* A square's jig has 4 arms in 2D, and its relation space is two-dimensional ($w_0 + w_2 = 0$ and $w_1 + w_3 = 0$). So $V = 0$ forces only $r_0 = r_2$ and $r_1 = r_3$: a rhombus, not a square. The missing equation is equal diagonals, which needs the rotation as well. That is the IVT-over-rotation step, and it is where continuity fails for non-convex curves. The same happens for the cross-polytope in every dimension. For simplices nothing is missing: the centring step alone forces regularity.

Theorem 3 — every convex body

Let $K \subset \mathbb{R}^n$ be a convex body (compact, convex, non-empty interior), not necessarily smooth or strictly convex. Let Δ, w_k and λ_k be as in Theorem 2, and for $Q \in O(n)$ write $u_k = Q w_k$. The **tangent cone** of K at $x \in \partial K$ is $T_K(x) = \overline{\bigcup_{t>0} t(K - x)}$, the directions into K from x.

> **Definition.** The orientation Q is **admissible** for K if at every boundary point $x \in \partial K$, at least one arm direction u_k lies in the interior of $T_K(x)$: from every boundary point, at least one arm of the jig points strictly into the body.

Theorem 3. If Q is admissible for K, then K contains an inscribed copy of Δ with orientation Q: there are $x^* \in K$ and $\rho > 0$ with $x^* + \rho\, u_k \in \partial K$ for every k.

Proof. **Step 1: smooth outer approximations.** By Schneider (*Convex Bodies*, Thm. 3.4.1), there are convex bodies L_m with C^∞ boundary and positive curvature, hence strictly convex, with Hausdorff distance $d_H(L_m, K + \frac{2}{m}B) \le \frac{1}{m}$, where B is the unit ball. Then $K + \frac{2}{m}B \subset L_m + \frac{1}{m}B$, i.e. $h_K + \frac{2}{m} \le h_{L_m} + \frac{1}{m}$ for the support functions, so $K + \frac{1}{m}B \subset L_m$. Also $L_m \subset K + \frac{3}{m}B$. Put $K_m = L_m$:

$$K \subset K_m \subset K + \tfrac{3}{m}B.$$

Step 2: Theorem 2 on each K_m. It gives $x_m \in \operatorname{int} K_m$ and $\rho_m > 0$ with $x_m + \rho_m u_k \in \partial K_m$. Because K_m is convex and x_m interior, each ray from x_m leaves K_m exactly once, so every arm of the jig at x_m has length exactly ρ_m.

Step 3: the simplices cannot shrink. Suppose $\rho_{m_j} \to 0$ along a subsequence. Passing to a further subsequence, $x_{m_j} \to x^*$, and $x^* \in K$ since $x_m \in K + \frac{3}{m}B$.

- *If $x^* \in \operatorname{int} K$:* some ball $B(x^*, 2\eta) \subset K \subset K_{m_j}$. For large j, $|x_{m_j} - x^*| < \eta$, so every arm at x_{m_j} has length at least η. This contradicts $\rho_{m_j} \to 0$.

- *If $x^* \in \partial K$:* by admissibility some $u_j \in \operatorname{int} T_K(x^*)$. The cone $C = \bigcup_{t>0} t(K - x^*)$ is convex with closure $T_K(x^*)$, so $\operatorname{int} T_K(x^*) = \operatorname{int} C$. Choose points $c_0, \dots, c_n \in C$ with u_j in the interior of their convex hull, and write $c_i = t_i(k_i - x^*)$ with $k_i \in K$, $t_i > 0$. For $0 < s \le s_0 := \min_i 1/t_i$, convexity gives $x^* + s c_i \in K$, hence $x^* + s \operatorname{conv}\{c_i\} \subset K$. So some ball $x^* + s_0 u_j + s_0 \eta B$ lies in K.

  Then every x with $|x - x^*| < s_0\eta$ has $x + s_0 u_j \in K \subset K_m$. For large j this holds for $x = x_{m_j}$, so the j-th arm at x_{m_j} has length at least s_0. Again this contradicts $\rho_{m_j} \to 0$.

So $\rho_m \ge c > 0$ for all large m; also $\rho_m \le \operatorname{diam} K_1$.

Step 4: the limit. Pass to a subsequence with $x_m \to x^*$ and $\rho_m \to \rho^* \in [c, \operatorname{diam} K_1]$. The vertices $y_k^m = x_m + \rho_m u_k \in \partial K_m$ converge to $y_k = x^* + \rho^* u_k$. Each $y_k \in K$, because $\partial K_m \subset K + \frac{3}{m}B$. And $y_k \notin \operatorname{int} K$: otherwise a ball around y_k lies in $K \subset K_m$, and for large m it contains y_k^m in the interior of K_m, contradicting $y_k^m \in \partial K_m$. So every $y_k \in \partial K$, and they are the vertices of the copy of Δ scaled by $\rho^* > 0$, centred at x^*, with orientation Q. ∎

Corollary 3.1. If ∂K is C^1, every orientation is admissible, so K contains an inscribed copy of Δ in every orientation. (Theorem 2 without strict convexity: flat pieces are fine.)

Proof. At $x \in \partial K$ with unit outer normal ν, $T_K(x) = \{d : \langle d, \nu\rangle \le 0\}$. By $(*)$, $\sum \lambda_k \langle u_k, \nu\rangle = 0$ with not all terms zero, so some $\langle u_k, \nu\rangle < 0$, i.e. $u_k \in \operatorname{int} T_K(x)$. ∎

> **Corollary 3.2 (polygons).** A convex polygon all of whose interior angles exceed 120° contains an inscribed equilateral triangle in every orientation.

Proof. At a vertex with interior angle $\beta > 120^\circ$, $\operatorname{int} T_K$ is an open sector of angle β, and any open arc longer than 120° contains one of three directions spaced 120° apart. Edge points are C^1 points. ■

(Regular polygons with at least seven sides qualify.)

> **Proposition 5 (admissibility cannot be dropped).** Let K be an isosceles triangle with apex angle $\alpha < 60^\circ$. Then K contains no inscribed equilateral triangle with a vertex pointing at the apex along the axis of symmetry.

Proof. Place the apex at the origin with the axis along the positive x-axis:
$K = \{(x, y) : 0 \le x \le 1,\ |y| \le hx\}$ with $h = \tan(\alpha/2) < 1/\sqrt{3}$. An equilateral triangle in this orientation, with circumcentre (c_x, c_y) and circumradius $\rho > 0$, has vertices $L = (c_x - \rho, c_y)$ and
$P_\pm = (c_x + \rho/2,\ c_y \pm \rho\sqrt{3}/2)$. If it were inscribed, all three vertices would lie in K, so the whole triangle would.

- $P_\pm \in K$ gives $|c_y| + \rho\sqrt{3}/2 \le h(c_x + \rho/2)$.
- $L \in \partial K$. It is not on the base, since then $P_\pm$ would have $x > 1$. So L is on a slanted side, the apex included: $|c_y| = h(c_x - \rho)$.

Substituting, $\rho\sqrt{3}/2 \le \frac{3}{2}h\rho$, i.e. $h \ge 1/\sqrt{3}$, a contradiction. ■

Remarks.

- Numerically, with the jig's arms at angles $\theta, \theta + 120^\circ, \theta + 240^\circ$ and the apex pointing in direction 180°, no inscribed equilateral triangle exists for any θ from 36° to 84° when the apex angle is 10°, or from 46° to 74° when it is 30°. Proposition 5 is the case $\theta = 60^\circ$, the middle of both ranges.
- The 60° bound is sharp: with apex 59° the orientation $\theta = 60^\circ$ still fails, and with apex 61° every orientation works. A right isosceles triangle also loses $\theta \in [46^\circ, 74^\circ]$, through its two 45° base corners. The thin rectangle, regular pentagon and regular heptagon lose none (`node tests/check-orientations.js`).
- This is the phenomenon Lemma B's C^1 assumption guards against. At a sharp corner the jig can be pinned with every arm pointing out of the body, and the would-be simplices shrink into the corner as the body is smoothed.
- *Some* orientation always works in the plane (Meyerson 1980: every Jordan curve has inscribed equilateral triangles). For general convex bodies in $\mathbb{R}^n$ an orientation-free version would follow from Theorem 3 if some orientation is admissible. That holds, for example, when no boundary point has a very sharp tangent cone.

Lemma 4 — a route to the open square problem

> **Lemma 4.** Let $\gamma : S^1 \to \mathbb{R}^2$ be a Jordan curve and $\gamma_L \to \gamma$ uniformly. Suppose there is $\delta \in (0, \frac{1}{2}]$ such that for infinitely many L, γ_L has an inscribed square with two opposite vertices $\gamma_L(u), \gamma_L(v)$ at parameter distance $|u - v|_{S^1} \geq \delta$. Then γ has an inscribed square.

Proof. Let $\eta = \min\{|\gamma(u) - \gamma(v)| : |u - v|_{S^1} \geq \delta\}$, which is positive because γ is injective and the set is compact. For large L, $\|\gamma_L - \gamma\|_\infty < \eta/4$, so those squares have diagonal at least $\eta/2$. Their vertex parameters lie in the compact $(S^1)^4$; pass to a convergent subsequence $u_i^L \to u_i$. Uniform convergence gives $\gamma_L(u_i^L) \to \gamma(u_i)$. A limit of squares whose diagonals are bounded below by $\eta/2$ is a non-degenerate square, so the four points $\gamma(u_i)$ form a square inscribed in γ. ■

How the experiments relate. For the fractal families in this repository, the level-L polygons, with their natural parametrisations, converge uniformly to the limiting fractal curve. The square shrinkage study finds, at every computed level, verified squares whose diagonal is at least about 0.8 of the curve's width, with their size settling as L grows. That is exactly Lemma 4's hypothesis, observed for finitely many L. A proof for a given fractal would need it for all L.

Why counting alone cannot do it. For generic smooth curves the number of inscribed squares is odd (Schnirelmann's argument: as the curve deforms, squares appear and disappear in pairs unless one shrinks to a point), while rectangles of any other aspect ratio come in even numbers. The square-count study confirms both on smooth and polygonal curves. On rough curves, small squares multiply at every scale and the count stops being meaningful, and in the limit it is infinite. So any proof for continuous curves must control the *sizes* of squares, as in Lemma 4, not just their number.

Theorem 6 — a computer-assisted certificate

Theorems 2 and 3 need smoothness or convexity. Lemma 4 needs control at *every* level of a fractal. This section takes a third route, which handles genuinely rough curves: prove that a square cannot be removed by any perturbation of a given size. Then compute, rigorously, that the size is large enough to cover a fractal.

Let E be the ellipse $x^2 + y^2/b^2 = 1$ with $b = 3/5$, with polar radius $r_E(\theta) = (\cos^2\theta + \sin^2\theta/b^2)^{-1/2}$. For a continuous $\rho : S^1 \to (0, \infty)$, the curve $\gamma_\rho = \{\rho(\theta)(\cos\theta, \sin\theta)\}$ is a Jordan curve: every ray from the origin crosses it once.

> **Theorem 6.** If ρ is continuous and $|\rho(\theta) - r_E(\theta)| \leq T = 0.15$ for all θ, then γ_ρ has an inscribed square, with diagonal at least 0.754.

The proof has a short analytic part, which holds for any base curve, and a finite computation in interval arithmetic (`node tools/certify-square.js` , under a second).

Parametrisation. Use the rational parametrisation of the ellipse, $\gamma_E(t) = \left(\frac{1-t^2}{1+t^2}, \frac{2bt}{1+t^2}\right)$, and parametrise the perturbed curve along the same rays: $\gamma_\rho(t) = \rho(\omega(t))\,\omega(t)$ with $\omega(t) = \gamma_E(t)/|\gamma_E(t)|$. Then $|\gamma_\rho(t) - \gamma_E(t)| = |\rho - r_E| \le T$ for every t. Opposite corners are $A = \gamma_\rho(t)$ and $C = \bar\gamma_\rho(t')$, where the second chart uses the opposite rays: $\bar\gamma_E(t') = -\gamma_E(t')$ (a point of E, which is centrally symmetric) and $\bar\gamma_\rho(t') = \rho(-\omega(t'))\,(-\omega(t'))$. Both charts are continuous and injective, and both satisfy the same bound T.

Square equations. For opposite corners A, C the other two corners of the square with diagonal AC are

$$B = \tfrac{1}{2}\big((I - R)A + (I + R)C\big), \qquad D = \tfrac{1}{2}\big((I + R)A + (I - R)C\big),$$

where R is the rotation by 90°. Let sd_ρ be the signed distance to γ_ρ (negative inside), and

$$G_\rho(t, t') = \big(\mathrm{sd}_\rho(B),\ \mathrm{sd}_\rho(D)\big) \in \mathbb{R}^2.$$

G_ρ is continuous, and $G_\rho = 0$ with $A \neq C$ is exactly an inscribed square.

Step 1 (perturbations move G by at most $(1+\sqrt{2})T$). Let $|\rho - \sigma| \le T$ pointwise.

- *Corners.* A and C move by at most T. The operators $\frac{1}{2}(I \pm R)$ have norm $1/\sqrt{2}$, so B and D move by at most $\sqrt{2}\,T$.
- *Signed distance.* For every point x, $|\mathrm{sd}_\rho(x) - \mathrm{sd}_\sigma(x)| \le T$. If x is on the same side of both curves, this holds because their Hausdorff distance is at most T. Otherwise write $x = s\,\omega$ with, say, $\sigma(\omega) \le s \le \rho(\omega)$. Then $|\mathrm{sd}_\rho(x)| + |\mathrm{sd}_\sigma(x)| \le (\rho(\omega) - s) + (s - \sigma(\omega)) \le T$, using the radial segment to each curve.
- *Together.* Since sd is 1-Lipschitz, $\|G_\rho - G_\sigma\|_\infty \le \sqrt{2}\,T + T$.

Step 2 (a degree lemma, in any dimension). Let $G_E, G_\rho : \mathbb{R}^n \supset Q \to \mathbb{R}^n$ be continuous with $\|G_\rho - G_E\|_\infty \le \tau$. Let $z \in \mathbb{R}^n$, let J and K be $n \times n$ matrices with $\det J \neq 0$, and put $Q = z + K([-c, c]^n)$ and $L(p) = J(p - z)$. On ∂Q, $L = JKy$ with $\|y\|_\infty = c$, so $\|L\|_\infty \ge c_{\text{eff}} := c\,(1 - \|JK - I\|_\infty)$. If

$$\|G_E - L\|_\infty + \tau < c_{\text{eff}} \quad \text{on } \partial Q,$$

then on ∂Q, $\|G_\rho - L\|_\infty < \|L\|_\infty$. So the homotopy $L + s(G_\rho - L)$, $s \in [0, 1]$, has no zero on ∂Q, and $\deg(G_\rho, Q, 0) = \deg(L, Q, 0) = \operatorname{sign}\det J \neq 0$. Hence G_ρ has a zero in Q.

A second sufficient condition allows larger boxes, because G_E may bend away from L as long as it keeps away from zero. Suppose that on ∂Q

$$\|G_E - L\|_\infty < \|G_E\|_\infty \qquad \text{and} \qquad \tau < \|G_E\|_\infty.$$

- If $(1 - s)L + sG_E = 0$ for some $s \in [0, 1)$, then $G_E - L = G_E/(1 - s)$ has norm at least $\|G_E\|$. So the first inequality makes the homotopy from L to G_E zero-free on ∂Q, and $\deg(G_E, Q, 0) = \operatorname{sign}\det J$. (This needs $c_{\text{eff}} > 0$, so that $L \neq 0$ on ∂Q.)
- The second inequality makes the homotopy from G_E to G_ρ zero-free, so $\deg(G_\rho, Q, 0) = \operatorname{sign}\det J \neq 0$.

The certificates check whichever condition holds; the theorems below use the second.

z, J and K can be any matrices: in practice an approximate zero, an approximate Jacobian and an approximate inverse. Their accuracy affects only how much margin is left, never correctness.

Step 3 (non-degeneracy). If $|\gamma_E(t) - \bar{\gamma}_E(t')| \geq \delta$ on Q, then the square from Step 2 has diagonal at least $\delta - 2T > 0$.

The computation (`pages/certify-rigorous.js`). Every quantity below is computed in interval arithmetic with outward rounding (`pages/interval.js` , see below), and only an approximate zero, J and K are plain floating-point data.

- *The box.* Newton's method finds the ellipse's square at $t = t' = 0.566190$. $Q = z + K([-c, c]^2)$ with $c = 0.5$; $\det J \in [0.9357, 0.9357]$, so the degree is $+1$, and $c_{\text{eff}} = 0.5000$ to nine places.
- *Exact signed distance.* sd_E is enclosed exactly, not via a polygon. The nearest point on the ellipse to p is $q_i = a_i^2 p_i/(t + a_i^2)$, where t is the unique root above $-b^2$ of the decreasing function $F(t) = \sum_i (a_i p_i/(t + a_i^2))^2 - 1$. A root found in floating point is enclosed in a bracket $[t_l, t_h]$ with $F(t_l) > 0 > F(t_h)$ verified in interval arithmetic. The distance $|t| \left(\sum_i p_i^2/(t + a_i^2)^2\right)^{1/2}$ is then enclosed over the bracket. At a box of possible points (the corners B, D are computed as interval boxes), sd is enclosed at a centre point and widened by the box's radius, because sd is 1-Lipschitz.
- *Lipschitz bound.* $\gamma_E'(t) = \big(-4t,\ 2b(1 - t^2)\big)/(1 + t^2)^2$, so $|\gamma_E'(t)| \leq 2/(1 + t^2) \leq 2$ for $b \leq 1$, and the same for $\bar{\gamma}_E$. With $|\Delta B| \leq (|\Delta A| + |\Delta C|)/\sqrt{2}$, G_E is $2\sqrt{2}$-Lipschitz in the max-norm on (t, t'), and $G_E - L$ is $(2\sqrt{2} + \|J\|_\infty)$-Lipschitz.
- *Sampling.* Each edge of $[-c, c]^2$ is sampled at 20,000 points (80,000 in all). Every point of ∂Q is within $\|K\|_\infty h/2$ of a sample, plus the floating-point error of computing the sample (bounded a priori by the standard γ_{2n} estimate). Between samples, G_E changes by at most 8.1×10^{-5} and $G_E - L$ by at most 1.1×10^{-4}. The enclosures give, on all of ∂Q:
  - $\|G_E\|_\infty \geq 0.3661 - 0.0001 = 0.3660 > (1 + \sqrt{2})\,T = 0.3621$;
  - $\|G_E\|_\infty - \|G_E - L\|_\infty \geq 0.2321 - 0.0002 > 0$, so the degree is $+1$.
- *Diagonal.* Covering Q's bounding box with 200×200 cells and evaluating $|\gamma_E(t) - \bar{\gamma}_E(t')|$ in intervals gives $\delta = 1.0541$, so the diagonal is at least $1.0541 - 0.30 = 0.754$.

Steps 1–3 then give Theorem 6. The same box certifies every $T \leq 0.1516$. (The first condition of Step 2 alone gives 0.1433.) Other ellipses do less well: $b = 1/2$ and $b = 7/10$ reach about 0.12, and $b = 4/5$ less than 0.08 . ■

Interval arithmetic and what is trusted. Each operation $+, -, \times, \div, \sqrt{}$ is computed with IEEE 754 round-to-nearest, which is within half a unit in the last place of the exact result. The result is then widened outward by $|x| \cdot 2^{-51} + 2^{-1074}$, which is at least one unit in the last place, so the exact result always lies inside. No other library functions are used (JavaScript does not specify the accuracy of `Math.sin` , `Math.cos` , …): the ellipse has a rational parametrisation, and the constants $3/5$, $4/5$ and T are enclosed in intervals because they are not binary fractions. What remains trusted is the JavaScript engine's IEEE arithmetic and the code itself (about 300 lines in `pages/interval.js` and `pages/certify-rigorous.js`), which is short enough to audit or port to a verified interval library.

Independent check. `tools/verify_certificates.py` re-implements every check from scratch in Python, using the interval arithmetic of the established library mpmath instead of `pages/interval.js`. It shares only the box definitions z, J, K, which the degree lemma allows to be arbitrary. It confirms Theorem 6 in full (the same 80,000 samples; $\|G_E\|_\infty \geq 0.3660 > 0.3621$, degree margin 0.2321, diagonal ≥ 0.75). It also confirms Theorem 7 in full, with the same adaptive cells: the same 997,168 cells, none failing, with the same margins and circumradius ≥ 0.4089. A third implementation, `tools/verify-certificates.c`, does the same in C with the ball arithmetic of Arb (FLINT 3). It reaches the same results, including the same 997,168 cells, in about 4 seconds.

Corollary 6.1 (fractal curves). Let $W(\theta) = (1 - 2^{-1/2}) \sum_{k \geq 0} 2^{-k/2} \cos\left(2^k(\theta - \theta_0)\right)$, so $|W| \leq 1$. Then the curve $\rho = r_E + 0.15\, W$ has an inscribed square, and so does $\rho = r_E + 0.15\, f$ for every continuous f with $|f| \leq 1$: random phases, Brownian-like noise, and so on.

For $\theta_0 = 0.3$ (the curve on the Square Certificate page), this curve is outside the classical cases:

- *Not smooth or Lipschitz.* W is nowhere differentiable, by Hardy's theorem, since $2 \cdot 2^{-1/2} \geq 1$.
- *Not locally monotone.* The curve is not locally a graph in any direction anywhere, so Stromquist's theorem does not apply. If a coordinate of γ_ρ were monotone on some interval of θ, it would be differentiable almost everywhere there (Lebesgue). Dividing by the smooth, non-vanishing factor $\cos(\theta - \varphi)$ away from one point would then make ρ, and so W, differentiable almost everywhere.
- *Not symmetric.* W is symmetric about θ_0 but not under $\theta \mapsto \theta + \pi$, and r_E is symmetric about 0 and $\pi/2$. So the curve has no reflection or point symmetry, and the symmetric-curve results of Nielsen and Wright do not apply.

Chambers (2022) proves that a Jordan curve close enough to a C^2 curve β has an inscribed square, with closeness measured against $1/\kappa$ for κ the maximum curvature of β. For $\beta = E$, $\kappa = 1/b^2 = 2.78$ and $1/(10\kappa) = 0.036$, so Theorem 6 covers a neighbourhood about 4.2 times wider. We have not ruled out that some other smooth β, with smaller curvature, lies within Chambers' bound of every curve in Corollary 6.1. So we claim only a computer-assisted proof of a quantitative statement, not a new class of curves.

What did not work: the Weierstrass blob. The method was also tried, in an earlier floating-point version (`tools/certify-float.js`), on the `Weierstrass` fractal of the finders, $\rho = 1 + \sum_k 0.085 \cdot 2^{-k/2} \cos(m 2^k \theta + \phi_k)$. The base curve is the K-octave truncation, and T is the size of the remaining tail, $\sum_{k \geq K} 0.085 \cdot 2^{-k/2}$:

K	tail to cover	best certifiable T	ratio
2	0.1451	0.0256	5.7
3	0.1026	0.0230	4.5
4	0.0726	0.0139	5.2

The ratio does not improve with K. Each added octave shrinks the tail by $2^{-1/2}$, but it also makes the base curve rougher at a finer scale, and the certifiable margin shrinks at the same rate. So the blob cannot be certified this way. (The exact signed distance used now gains little: 0.141 → 0.143 for the ellipse, far short of a factor of

5.) The ellipse works because a large, smooth base square survives a fixed-size perturbation; a curve whose roughness carries a constant fraction of its size at every scale does not offer one. This is the quantitative form of the difficulty in the open problem.

Theorem 7 — the same for tetrahedra on rough surfaces

The inscribed-tetrahedron problem is open exactly for non-smooth surfaces: Gromov's theorem needs C^1. Theorem 6's method extends to 3D if the tetrahedron's orientation is fixed, and the degree lemma of Step 2 applies unchanged with $n = 4$.

Let E be the ellipsoid $x^2 + y^2/(4/5)^2 + z^2/(3/5)^2 = 1$ with radial function r_E. For continuous $\rho : S^2 \to (0, \infty)$ let $S_\rho = \{\rho(\omega)\,\omega : \omega \in S^2\}$, a Jordan surface. For the orientation, take the exactly orthogonal rational rotation (Cayley transform)

$$R = \frac{(1 - |q|^2)I + 2qq^{\mathsf{T}} + 2[q]_\times}{1 + |q|^2}, \qquad q = (\tfrac{1}{8}, \tfrac{1}{4}, \tfrac{3}{8}),$$

and $w_k = Rv_k/\sqrt{3}$, where the v_k are $(\pm 1, \pm 1, \pm 1)$ with an even number of minus signs. These are the directions from the centre to the vertices of a regular tetrahedron.

> **Theorem 7.** If $|\rho(\omega) - r_E(\omega)| \le T = 0.13$ for all ω, then S_ρ has an inscribed regular tetrahedron with vertices $x + s\,w_k$, so in that orientation, with circumradius $s \ge 0.4089$.

Proof.

- **Equations.** The unknowns are $p = (x, s) \in \mathbb{R}^4$. Let $G_\rho(p) = \big(\mathrm{sd}_\rho(x + sw_k)\big)_{k=0..3}$. A zero with $s > 0$ is an inscribed tetrahedron. The vertices depend only on p, not on the surface, so by the radial argument of Theorem 6, Step 1 (which works verbatim in 3D), $\|G_\rho - G_E\|_\infty \le T$.
- **Box.** Newton's method gives the tetrahedron on E, with centre $(-0.2856, 0.0120, 0.0466)$ and circumradius 0.7158. $Q = z + K([-c, c]^4)$ with $c = 0.26$; $\det J \in [1.9345, 1.9345]$, so the degree of L is $+1$.
- **Lipschitz bound.** $|\Delta(x + sw_k)| \le |\Delta x|_2 + |\Delta s|\,|w_k| \le (\sqrt{3} + 1)\|\Delta p\|_\infty$, with $|w_k| \le 1$ up to rounding (enclosed), so G_E is $(\sqrt{3} + 1)$-Lipschitz and $G_E - L$ is $(\sqrt{3} + 1 + \|J\|_\infty)$-Lipschitz.
- **The computation** (`node tools/certify-tetra.js` , a few seconds on 4 cores), in the same interval arithmetic, with the same exact enclosure of the signed distance to the ellipsoid, in **adaptive cells**:
  - Each of the 8 faces of $[-c, c]^4$ starts as 8^3 cells. A cell is checked at its centre. Every point of its image in Q lies within $r = \|K\|_\infty h + \varepsilon$ of the centre's sample point, where h is the cell's half-width and ε the rounding in computing that point. So on the whole cell $\|G_E\|_\infty \ge \|G_E(\text{centre})\|_\infty - (\sqrt{3} + 1)r$, and $\|G_E\|_\infty - \|G_E - L\|_\infty$ is at least its value at the centre minus $(2(\sqrt{3} + 1) + \|J\|_\infty)r$.
  - A cell passes if the first bound exceeds T and the second is positive. Otherwise it is split in half in every direction (exact in floating point, so the cells always cover the face exactly).

    - 997,168 cells were checked, splitting at most 6 times, and every one passes. So on all of ∂Q, $\|G_E\|_\infty > T = 0.13$ and $\|G_E - L\|_\infty < \|G_E\|_\infty$, and by Step 2 of Theorem 6 (second condition) the degree is $+1$ for every surface within T. The same box certifies $T = 0.134$.
    - With a uniform grid instead, every part of the boundary would need the fine spacing that only the tightest spots need: 64 million samples for the weaker $T = 0.12$.
- **Non-degeneracy.** s is affine on Q, so its minimum, 0.4089, is attained at one of the 16 corners (enclosed in intervals). Its maximum is 1.0228. ■

Corollary 7.1 (rough surfaces). Let $F : S^2 \to [-1, 1]$ be any continuous function. For example, take $F(\omega) = (1 - 2^{-1/2}) \sum_{k \geq 0} 2^{-k/2} \cos(2^k \langle \omega, e \rangle)$ for a fixed unit vector e, which is nowhere differentiable along every meridian through e (Hardy). Then the surface $\rho = r_E + 0.13\, F$ has an inscribed regular tetrahedron. Such surfaces are not C^1, or even Lipschitz, so Gromov's theorem does not apply. We know of no published result that covers them, but, as for Theorem 6, the claim is a quantitative computer-assisted one and not a new structural class.

Corollary 7.2 (an open set of orientations). Turning the tetrahedron by an angle θ moves each vertex $x + s\, w_k$ by $s\, |R' w_k - R w_k| \leq s\, \theta$, so it moves G by at most $s_{\max} \theta$, with $s_{\max} = 1.0228$ on Q. That is one more perturbation, so certifying the box for the allowance $T + s_{\max} \theta$ gives, for every surface within T of E, an inscribed regular tetrahedron in **every** orientation within angle θ of the given one. The largest θ certified (by bisection, each step a full adaptive certificate):

T	**0**	**0.03**	**0.06**	**0.09**	**0.12**
orientations within	7.52°	5.84°	4.16°	2.48°	0.81°

Corollary 7.3 (every orientation). If $|\rho(\omega) - r_E(\omega)| \leq T = \frac{7}{100}$ for all ω, then S_ρ has an inscribed regular tetrahedron in **every** orientation.

This is the analogue, for these rough surfaces, of Gromov's "every pose" theorem for C^1 spheres.

Proof.

- **Orientations as quaternions.** Every rotation is $R(x)$ for a unit quaternion $x = (w, v)$, determined up to sign. When $w \neq 0$, the Cayley vector $q = v/w$ gives $R(x)$ as the matrix in Theorem 7. The tetrahedron's 12 rotational symmetries are $R(u)$ for the 24 Hurwitz units $u \in \{\pm 1, \pm i, \pm j, \pm k, \frac{1}{2}(\pm 1 \pm i \pm j \pm k)\}$. For example, i is the half-turn $(x, y, z) \mapsto (x, -y, -z)$ and $\frac{1}{2}(1 + i + j + k)$ permutes the coordinates cyclically, and each permutes the v_k. So $R(x)$ and $R(xu) = R(x)R(u)$ give the same four directions w_k, in a different order, and it is enough to prove the result for one xu.
- **A fundamental domain.** Given x, choose a unit u that maximises $\langle x, u \rangle$ and put $y = x\bar{u}$, so $\bar{u}$ is again a unit. Right multiplication by a unit quaternion is an isometry that permutes the units. So $\langle y, 1 \rangle = \langle x, u \rangle \geq \langle x, u'u \rangle = \langle y, u' \rangle$ for every unit u'. Write $y = (w, v)$. The units $\frac{1}{2}(1 \pm i \pm j \pm k)$ give $w \geq |v_1| + |v_2| + |v_3|$, so $w > 0$ and the Cayley vector of y lies in the solid octahedron $\Omega = \{|a| + |b| + |c| \leq 1\}$. It is therefore enough to cover Ω.

- **Turning within a cube.** Let q, q' lie in a cube of half-width h, and let u, u' be the unit quaternions $(1,q)/|(1,q)|$ and $(1,q')/|(1,q')|$.
  - $|u'-u| \le |q'-q|/d \le \sqrt{3}\,h/d$, where d is the smallest value of $|(1,q)|$ on the cube, because radial projection onto the unit sphere is $(1/d)$-Lipschitz on points at distance at least d from the origin. Away from $q=0$, $d>1$, which helps.
  - For a unit vector v, $|R(u')v - R(u)v| = |(u'-u)v\bar{u}' + uv(\bar{u}'-\bar{u})| \le 2|u'-u|$.
  - So each vertex $x + s\,w_k$ moves by at most τs, with $\tau = 2\sqrt{3}\,h/d$, and since signed distance is 1-Lipschitz, G moves by at most τs at a point of Q with circumradius s.
  - As in Corollary 7.2, certifying orientation q with the allowance $T + \tau s$ proves the result for every orientation in the cube. The allowance is bounded cell by cell, with s replaced by its largest value on the cell, rather than by one bound $T + \tau s_{\max}$ for the whole box.
- **The computation** (`node tools/certify-orientations.js --T 7/100`, about 4 hours on 4 cores).
  - $[-1,1]^3$ is split into 36^3 cubes, of half-width $h = 1/36$, and the 10,544 that meet Ω are kept.
  - For each one, Newton's method from a fixed start finds the tetrahedron on E in the orientation of the cube's centre, and a box with $c = 0.24$ is built around it. If that box fails, the program tries $c = 0.22$ and 0.20, and if those fail too it splits the cube into 8 cubes of half the width and certifies each of them in the same way.
  - The box is certified as in Theorem 7, with adaptive cells and the allowance $T + \tau s$, where $\tau \le 0.0963$.
  - 10,224 cubes pass with $c = 0.24$. The other 320 fail with all three boxes and are split; all 2,560 halves pass with $c = 0.24$ (their τ is half as large). So 12,784 certificates cover Ω, with 16.3 billion cells in all, and the circumradius is at least 0.4239 throughout.
  - Cells are split only until they pass, so on the tightest cells the margin is small (the smallest is 2.4×10^{-9}). It is still a strict inequality in outward-rounded interval arithmetic, which is all the degree lemma needs.
  - The results are in `data/orientations-progress-T7-100.jsonl` (one line per cube, including the 320 that were split, each recording the T it was certified for) and `data/certificate-orientations-T7-100.json`. ■

Checks.

- `python3 tools/verify_certificates.py orientations --T 7/100` re-checks the covering in exact rational arithmetic, without the JavaScript's bookkeeping. The coordinates bounding the certified cubes form a grid, and every cell of that grid that meets Ω lies inside a certified cube.
- `./verify orientations data/orientation-sample-T7-100.json` re-checks a seeded random sample of 100 of the 12,784 certified cubes (`node tools/certify-orientations.js --T 7/100 --sample 100`; 24 of them are halves of split cubes) in C with Arb. It takes only T, each cube's corners, its centre and the box data z, J, K, and recomputes the rotation, τ, the circumradius range and every cell itself. All 100 pass: 137 million cells, about 6 minutes.
- Re-checking all 12,784 cubes in Arb would take about 13 hours on 4 cores. It has not been done.

How far does this go? An earlier version charged the whole box the largest turning cost $2\sqrt{3}\,h\,s_{\max} \approx 0.093$ and reached only $T = 0.02$. Charging each cell its own τs, and using d, saves about 0.03 where the room is tightest, and the larger box $c = 0.24$ gives more room for a general orientation. At $T = 0.05$ every cube of half-width $1/36$ passes, with at least 0.0079 to spare. At $T = 0.07$, 320 of them (3%) need splitting, and the margins become small, so going further would mean splitting many more cubes, at about 8 times the cost per halving of h.

Remarks.

- The degree lemma needs no winding numbers and works in any number of unknowns. It only needs the equations to stay away from zero, and either close to their linearisation or closer to it than to zero, on the boundary of a box.
- The neighbourhood is smaller than in 2D (0.13 against 0.15) because the tetrahedron equations are more curved relative to the box. At $c = 0.28$ the degree condition fails, so $c = 0.26$ is about as far as one box goes.

What remains open

Problem	Status
Square in every continuous Jordan curve (Toeplitz)	**Open.** Proved for convex (Emch), smooth (Schnirelmann), locally monotone (Stromquist) and small-Lipschitz (Tao) curves
Rectangle of every aspect ratio in every continuous Jordan curve	**Open.** Proved for smooth curves (Greene–Lobb) and extended to rectifiable curves; *some* rectangle always exists (Vaughan)
Regular tetrahedron / simplex in every continuous Jordan sphere	**Open.** Proved for C^1 spheres in every pose (Gromov 1969); Theorems 2–3 give elementary proofs for every C^1 convex body and, in admissible orientations, every convex body. Theorem 7 certifies every surface within 0.13 of one ellipsoid, and Corollary 7.3 every orientation for every surface within 0.07. "Every pose" fails for non-smooth bodies (Proposition 5)
Regular octahedron / cross-polytope	**Open.** Proved for smooth convex bodies in $\mathbb{R}^3$ (Makeev 2001) and in $\mathbb{R}^n$ for n an odd prime power (Karasev 2009)
Cube / hypercube ($n \geq 3$)	**False in general** (Theorem 1), with an explicit strictly convex surface that has no inscribed cube (Theorem 1.3). True for centrally symmetric convex bodies in $\mathbb{R}^3$ (Hausel–Makai–Szűcs 2000)

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